WORKED SOLUTION

STEP II 2001, Question 5

Pure • Stationary Points and Integration

STEP II 2001, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s2-q5

The key idea

Stationary points come straight from the gradient: they sit where $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$, and their nature is decided by how the sign of the gradient changes. Because the exponential factor is always positive, that sign depends only on the polynomial factor $x(1-x^2)$, which is the same for both curves. For the heights, the first gradient integrates exactly (spot $u = x^2$); the second doesn't, but we don't need its value, only a comparison of two integrals.

Solution

The curve C_1

Since $\mathrm{e}^{-x^2} > 0$ for every x, the gradient is zero exactly when $x(1-x^2) = x(1-x)(1+x) = 0$, that is, at $x = -1$, 0 and 1. Also the gradient has the same sign as $x(1-x)(1+x)$:

	$x < -1$	$-1 < x < 0$	$0 < x < 1$	$x > 1$
sign of $\frac{\mathrm{d}y}{\mathrm{d}x}$	$+$	$-$	$+$	$-$

At $x = 0$ the gradient changes from negative to positive, so the origin (which is on the curve) is a minimum. At $x = 1$ it changes from positive to negative, so this is a maximum; likewise $x = -1$ is a maximum.

To find the heights we integrate. Since the curve passes through the origin,

$$y = \int_0^x t(1-t^2)\,\mathrm{e}^{-t^2}\,\mathrm{d}t.$$

? **Ask yourself** How do we integrate $t(1-t^2)\mathrm{e}^{-t^2}$?

The factor t next to a function of t^2 suggests $u = t^2$, $\mathrm{d}u = 2t\,\mathrm{d}t$:

$$y = \tfrac{1}{2}\int_0^{x^2} (1-u)\,\mathrm{e}^{-u}\,\mathrm{d}u.$$

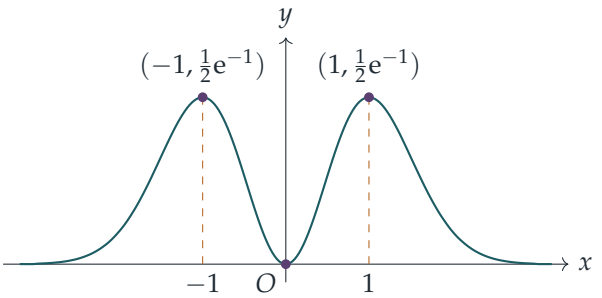
Now notice that $\frac{\mathrm{d}}{\mathrm{d}u}(u\mathrm{e}^{-u}) = \mathrm{e}^{-u} - u\mathrm{e}^{-u} = (1-u)\mathrm{e}^{-u}$ (integration by parts gives the same thing). Hence

$$y = \tfrac{1}{2}\Big[u\,\mathrm{e}^{-u}\Big]_0^{x^2} = \tfrac{1}{2}\,x^2\mathrm{e}^{-x^2}.$$

At $x = 1$ this gives $y = \frac{1}{2}\mathrm{e}^{-1}$, so the maximum is at $(1, \frac{1}{2}\mathrm{e}^{-1})$, as required. Since y is an even function of x, the other stationary point is the maximum

$$(-1, \tfrac{1}{2}\mathrm{e}^{-1}).$$

Sketch. From $y = \frac{1}{2}x^2\mathrm{e}^{-x^2}$: the curve is symmetric about the y-axis, $y \geq 0$ with equality only at the origin, and $y \to 0$ as $x \to \pm\infty$ because the exponential decays faster than x^2 grows. So it is a "double hump": it touches the x-axis at the origin, rises to two equal maxima at $x = \pm 1$ (height $\frac{1}{2}\mathrm{e}^{-1} \approx 0.18$), then dies away towards the x-axis on both sides. (Vertical scale exaggerated below.)



The curve C_2

Again $\mathrm{e}^{-x^3} > 0$ for every x, so the gradient vanishes exactly at $x = -1, 0, 1$ and has the same sign as $x(1-x)(1+x)$, giving exactly the same sign table as before. So the origin (on the curve by hypothesis) is a minimum, and the stationary point with $x = 1$ is a maximum. (The one at $x = -1$ is a maximum too, though we aren't asked about it.)

Its height is

$$k = \int_0^1 x(1-x^2)\,\mathrm{e}^{-x^3}\,\mathrm{d}x.$$

This has no elementary antiderivative, so instead we compare it with the C_1 integral, which we know equals $\frac{1}{2}\mathrm{e}^{-1}$.

? **Ask yourself** How can we compare k with $\frac{1}{2}\mathrm{e}^{-1}$ without evaluating k?

Compare the integrands. For $0 < x < 1$ we have $x^3 < x^2$ (multiply $x < 1$ by $x^2 > 0$), so $-x^3 > -x^2$ and, as e^t is increasing, $\mathrm{e}^{-x^3} > \mathrm{e}^{-x^2}$. Also $x(1-x^2) > 0$ there, so multiplying preserves the inequality:

$$x(1-x^2)\,\mathrm{e}^{-x^3} > x(1-x^2)\,\mathrm{e}^{-x^2} \qquad \text{for } 0 < x < 1.$$

Both integrands are continuous on $[0,1]$ and the inequality is strict on the open interval, so integrating over $[0,1]$ gives a strict inequality:

$$k = \int_0^1 x(1-x^2)\,\mathrm{e}^{-x^3}\,\mathrm{d}x > \int_0^1 x(1-x^2)\,\mathrm{e}^{-x^2}\,\mathrm{d}x = \tfrac{1}{2}\mathrm{e}^{-1},$$

the last equality being the height of C_1 at $x = 1$ found above. So the maximum is at $(1,k)$ with $k > \frac{1}{2}\mathrm{e}^{-1}$, as required.

Checks

Differentiating $\frac{1}{2}x^2\mathrm{e}^{-x^2}$ by the product rule gives $x\mathrm{e}^{-x^2} - x^3\mathrm{e}^{-x^2} = x(1-x^2)\mathrm{e}^{-x^2}$, and it is 0 at $x = 0$, so C_1 is right. Second derivatives confirm the classification: for both curves $y''(0) = 1 > 0$ and $y''(1) = -2\mathrm{e}^{-1} < 0$. Numerically (Simpson's rule), $k \approx 0.2034$, while $\frac{1}{2}\mathrm{e}^{-1} \approx 0.1839$.

Common slips

- Forgetting the constant of integration: the condition "passes through the origin" is what fixes it, and it matters for the heights.
- Claiming $k > \frac{1}{2}\mathrm{e}^{-1}$ because $\mathrm{e}^{-x^3} > \mathrm{e}^{-x^2}$ without noting that $x(1-x^2)$ is positive on $(0,1)$; multiplying by a negative factor would reverse the inequality.
- Saying $x^3 < x^2$ "for all x": it is true only for $x < 1$, $x \neq 0$, which is why the interval $[0,1]$ is exactly what we need.

Beyond STEP

The second part is a comparison argument: we learn something about a solution we can't write down by comparing it with one we can. This idea runs right through the theory of differential equations, where comparison theorems bound solutions of awkward equations by solutions of simpler ones, and through analysis, where monotonicity of the integral is a basic tool. Boyce and DiPrima, *Elementary Differential Equations*, is a good place to see how much can be said about solutions qualitatively, without formulae.

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